Analysis of Commercial Trench Power MOSFETs' Responses to Co^{60} Irradiation

Sandra Liu, *Member, IEEE*, Christopher DiCienzo, Martin Bliss, Max Zafrani, *Member, IEEE*, Milton Boden, *Member, IEEE*, and Jeffrey L. Titus, *Senior Member, IEEE*

***Abstract*—This paper presents a detailed analysis of commercial trench power MOSFET responses to Co^{60} irradiation of all key d.c. electrical test parameters. Charge trapping in the gate oxide causes the threshold voltage ($V_{\rm TH}$) to shift, which, in turn, accounts for most of the degradation exhibited by the device after irradiation.**

***Index Terms*—Total ionizing dose (TID), trench power MOSFET.**

I. Introduction

THERE has been an increasing interest and effort to evaluate the use of commercial power trench devices for potential space applications. Most studies have focused only upon the devices' SEE performance and threshold voltage shifts [1]–[3], but have failed to examine how each electrical test parameter is affected by ionizing radiation under different bias conditions. For one to determine if commercial trench devices are acceptable for space applications and their limitations, it is desirable to fully understand how each electrical test parameter is affected by ionizing radiation and what caused those effects.

Fig. 1 is a schematic representation of a typical trench power MOSFET structure as presented in [4]. In contrast to radiation hardened power MOSFETs, most of which are planar DMOSFETs, trench MOSFETs are fabricated with trenches etched into the silicon; gate oxides are then grown along the trench sidewalls; and then the trenches are filled with polysilicon, which then becomes the gate. Incorporating a trench structure significantly increases the channel width when compared to the planar power DMOSFET, and also eliminates the JFET region (commonly referred to as the neck region in the vertical power MOSFET), allowing trench devices to conduct higher current densities and to exhibit lower on-resistances [4], [5]. In some trench technologies, a thicker oxide is used at the trench bottom to dramatically reduce the capacitance between the gate and drain and to enhance the device's switching performance [4]. Given these advantages, it is not surprising that designers want to use trench power MOSFETs for use in potential space

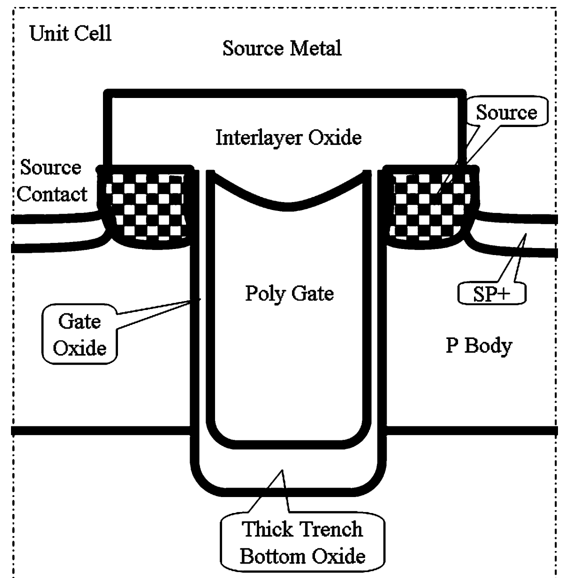

Fig. 1. A schematic of a typical N channel trench power MOSFET with thick trench bottom oxide.

application. However, before these commercial trench devices can be used, they must be capable of operating in a space environment, which requires them to be characterized to different types of radiation such as cosmic (heavy ions) and ionizing radiation (Co^{60}). Of course, a proper evaluation is critical to understand the device's true radiation capability and to provide meaningful recommendations for its intended applications. Trench power MOSFET has a unique cell structure when compared to conventional planar DMOSFETs. Thus, trench power MOSFETs present unusual challenges to test engineers. One of those challenges is how to properly test and interpret the performance of the trench MOSFET to various radiation stimuli such as heavy ions and total ionizing dose. Two recent studies show that trench power MOSFETs exhibit higher sensitivity to heavy ion irradiation due to its unique structure [6], [7].

This paper presents detailed total ionizing dose test results of advanced low-voltage commercial trench power MOSFETs manufactured by International Rectifier (IR). Total ionizing dose tests using a Co^{60} irradiation source are conducted. Devices from several different trench product lines are irradiated at different drain and gate biases up to dose levels of 50 krad(Si) using step irradiations. Key electrical parameters are measured after each radiation step.

Analyses and explanations are given on why some key electrical parameter shifts occurred under certain bias conditions

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S. Liu and M. Boden are with the International Rectifier Corporation, El Segundo, CA 90245 USA (e-mail: sliu1@irf.com; mboden1@irf.com).

C. DiCienzo, M. Bliss, and M. Zafrani are with the International Rectifier Corporation, Leominster, MA 01254 USA (e-mail: cdicienz1@irf.com; mbliss11@irf.com; mzafran1@irf.com).

J. L. Titus is with the Crane Division, NAVSEA, Crane IN 47522 USA (e-mail: Jeffrey.Titus@navy.mil).

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after irradiation. Most of these shifts can be attributed to large threshold voltage shifts caused by charge trapping in the gate oxide. The large threshold voltage shift is mainly due to commercial trench power MOSFETs using unhardened gate oxides.

II. Sample Info and Test Setup

The trench power MOSFETs selected for this study are n-channel MOSFETs from IR's most popular trench product families. Three device types are chosen to include three drain blocking voltages rated at 30 V, 40 V, and 75 V. All devices have rated gate voltages of ± 20 V, but not necessarily the same gate oxide thickness. The 30 VN device uses a nominal 450 Å gate oxide thickness; and, the 40 VN and 75 VN devices use a nominal 750 Å gate oxide thickness. All three devices are designed using a stripe trench topology and exhibit extremely low on-resistance for their rated voltages.

For total ionizing dose tests, die are assembled and mounted into regular TO-3 type packages. All samples are electrically tested and verified as functional before shipping them to the irradiation site at the University of Massachusetts (Lowell). The radiation source used for these tests is a Co^{60} source and the average dose rate for our exposures is approximately 107 rad(Si)/s. Parts are irradiated with different in-situ biases. In-situ biases, consisting of four fixed drain biases (e.g., the drain biases for 30 VN device are 0 V, 8 V, 16 V, and 24 V) or five fixed gate biases (selected gate biases are -16 V, -8 V, 0 V, 8 V, and 16 V) and the other terminals are grounded. Step irradiations are performed to acquire electrical test data after total ionizing dose levels of 2, 5, 10, 25, and 50 krad(Si). After each step irradiation, devices are electrically characterized within 15 minutes of the irradiation step. For each in-situ bias condition, a minimum of two samples are characterized.

Since all three device types exhibited similar behavior, we only show a portion of the actual test results and most of the illustrations used in this paper are for the 30 V N-channel device.

III. Experimental Results & Discussions

A. *Effects of In-Situ Gate and Drain Biases Upon Threshold Voltage* (V_{TH})

Commercial trench power MOSFETs usually use thinner gate oxides than their radiation-hardened planar power DMOSFET counterparts. However, commercial trench gate oxides tend to trap more positive charges due to the process differences such as how the gate oxide is grown, and what post-gate thermal cycles the gate oxide experiences after it is grown, which, in turn, generates much more charge trap centers and induces larger V_{TH} shifts [8]. Therefore, it is somewhat expected that the threshold voltage shifts of commercial trench power MOSFETs should exhibit large shifts under total ionization dose test. Figs. 2–4 show that the V_{TH} decreases linearly with increasing dose for the five different gate biases used during irradiation. In addition, these data show that the largest V_{TH} shift occurs when the in-situ gate bias is 16 V. Based on these plots, V_{TH} shifts to below zero volts at approximately 29 krad(Si) for the 30 VN and 40 VN devices and at approximately 27 krad(Si) for the 75 VN device. If V_{TH} becomes less than zero volts, the device changes from an enhancement mode (normally off) device to a depletion mode(normally on) device, which then requires a negative gate

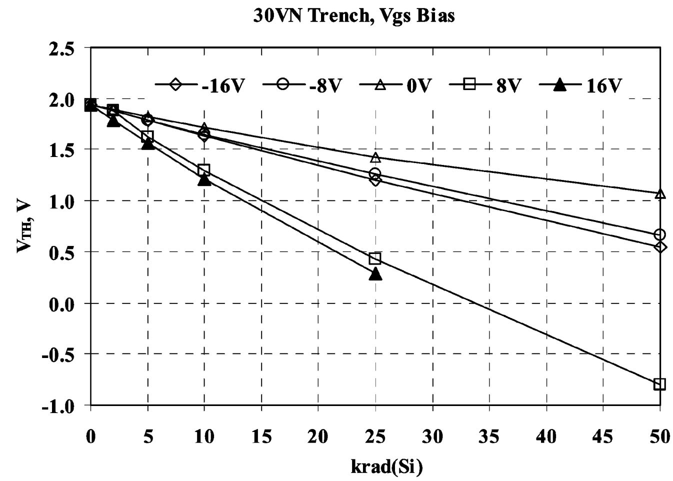

Fig. 2. V_{TH} shifts of 30 VN trench power MOSFETs vs. ionizing radiation dose under different in-situ gate biases.

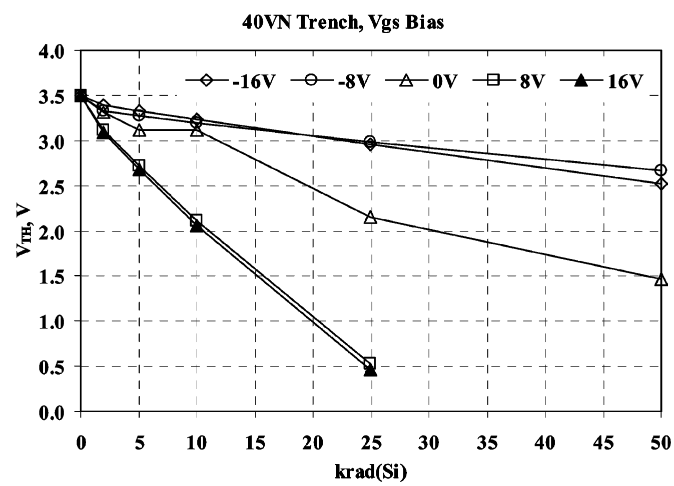

Fig. 3. V_{TH} shifts of 40 VN trench power MOSFETs vs. ionizing radiation dose under different in-situ gate biases.

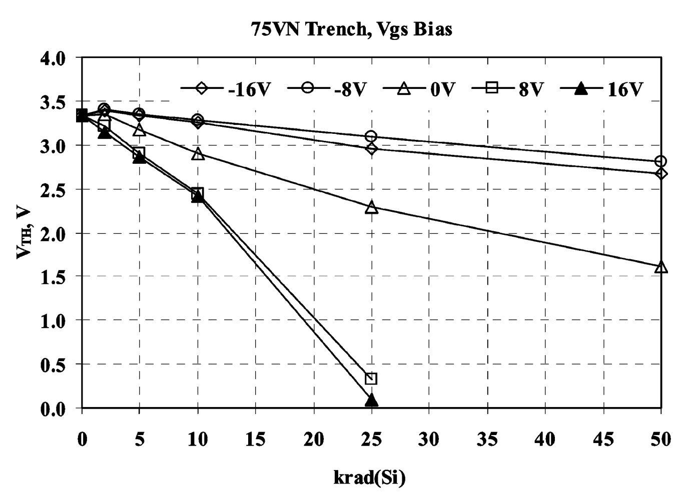

Fig. 4. V_{TH} shifts of 75 VN trench power MOSFETs vs. ionizing radiation dose under different in-situ gate biases.

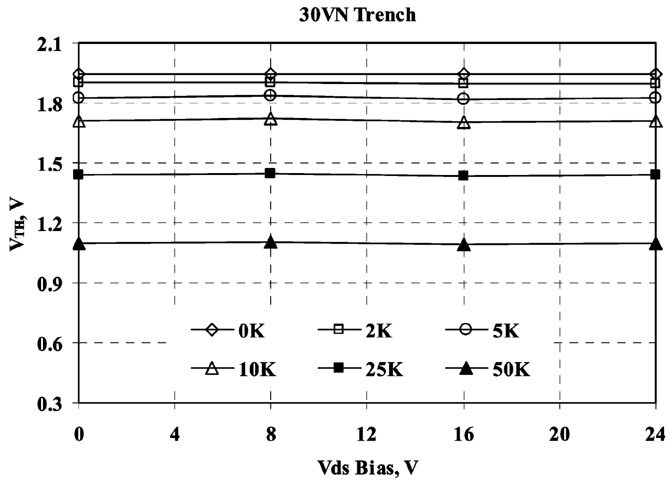

Fig. 5. At all radiation levels tested, in-situ drain bias has little impact upon V_{TH} shift. Although V_{TH} decreases linearly with increasing radiation levels, these changes are similar to the changes as with in-situ gate bias of zero volts (see Fig. 2).

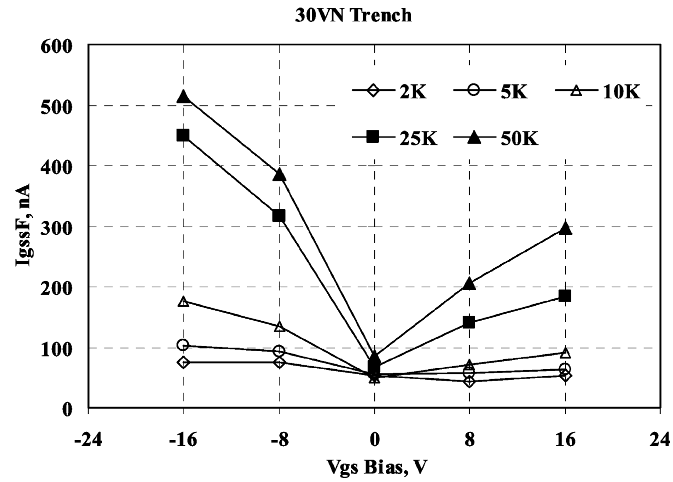

Fig. 7. The impact of gate bias voltage under radiation on forward-biased gate leakage current (IgssF) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

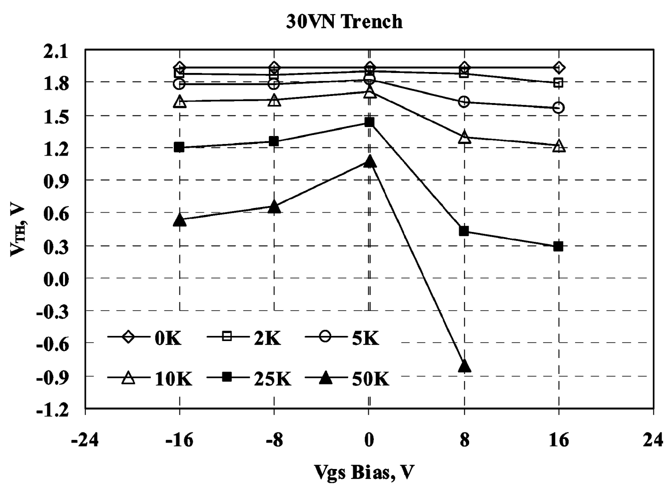

Fig. 6. Another way of looking at the effect of gate bias on the threshold voltage shift relative to total ionizing dose levels. Threshold voltage decreased with both positive and negative bias under irradiation, but much larger shifts are observed with more positive gate biases at higher dose levels (30 VN).

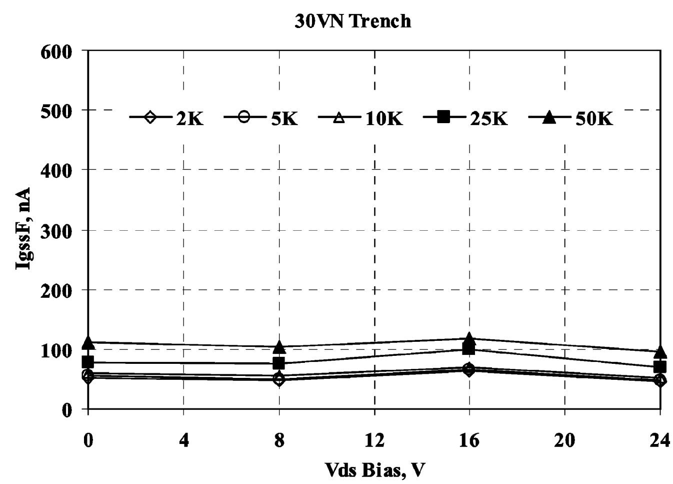

Fig. 8. The impact of drain bias voltage under radiation on forward-biased gate leakage current (IgssF) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

bias to turn the device off, which may not be practical from an application viewpoint.

Fig. 5 shows the V_{TH} responses of the 30 VN devices to different in-situ drain biases and dose levels. These data suggest that the in-situ drain bias has little impact upon V_{TH}. Although V_{TH} decreases with increasing dose, it is the same changes as devices tested at 0 V gate bias.

Fig. 6 is a plot showing how gradual V_{TH} changes with increasing dose under the different in-situ biases. The data plotted in Fig. 6 are the same data plotted in Fig. 2 but viewed in a different way. The V_{TH} shifts for each in-situ gate bias increases differently at each dose level with the largest shifts occurring for positive gate bias at 25 and 50 krad(Si).

Clearly, in-situ gate biases caused significant changes in I_{DSS} with increasing radiation doses. This effect is observed because under these in-situ biases, V_{TH} decreased significantly and in some cases shifted below zero volts. These shifts in V_{TH} would make the device either very leaky or become normally on. Under both of these situations, I_{DSS} increases significantly.

This observed trend of V_{TH} shifts under different gate biases for the 30 VN commercial trench MOSFET is similar to that observed in radiation-hardened power MOSFETs, except the magnitude of the V_{TH} shifts are smaller in planar radiation-hardened MOSFETs (approximately 3 times smaller when comparing drain bias effects and 14 times smaller when comparing gate bias effects per unit gate oxide thickness).

B. Effects of In-Situ Gate and Drain Biases on Gate Leakage Currents (I_{GSS})

The gate leakage current of a MOSFET is typically tested at two bias conditions: (1) when gate is positively biased at rated gate voltage—such as 20 V; and (2) when gate is negatively biased at rated gate voltage—such as -20 V; while drain is grounded. For an N channel MOSFET, the leakage current at positive gate bias is called forward-biased gate leakage current—IgssF, and the leakage current at negative gate bias is

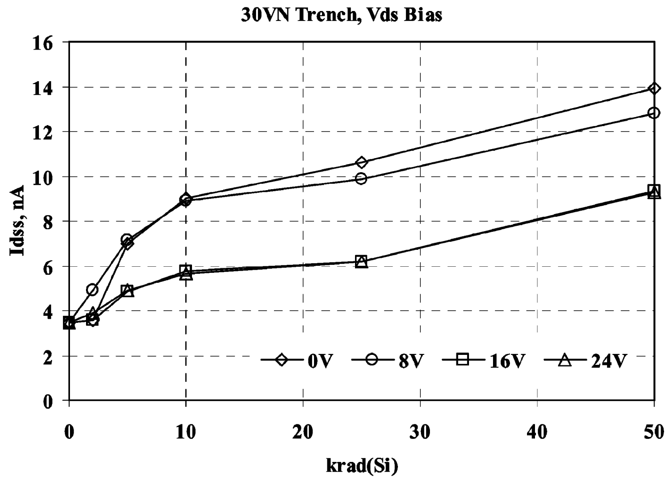

Fig. 9. Changes in $I_{\rm DSS}$ with increasing radiation doses under different in-situ drain biases for a 30 V N-channel trench power MOSFET.

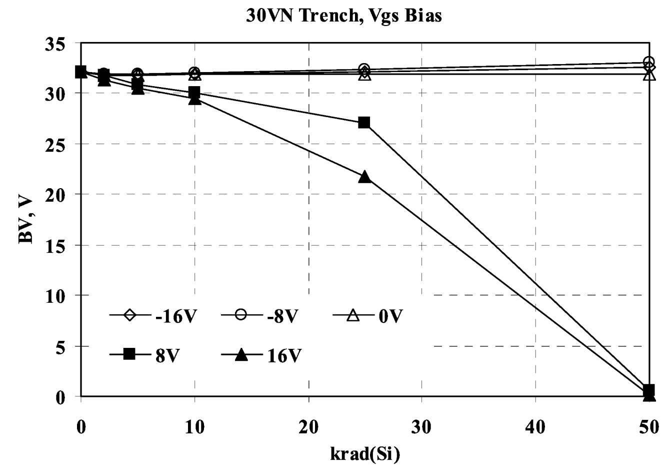

Fig. 11. The impact of gate bias voltage under radiation on breakdown voltage (BV) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

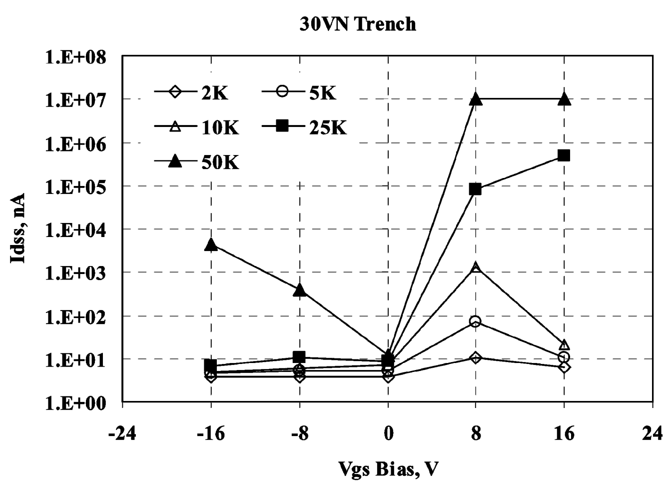

Fig. 10. Changes in $I_{\rm DSS}$ with increasing radiation doses under different in-situ gate biases for a 30 V N-channel trench power MOSFET.

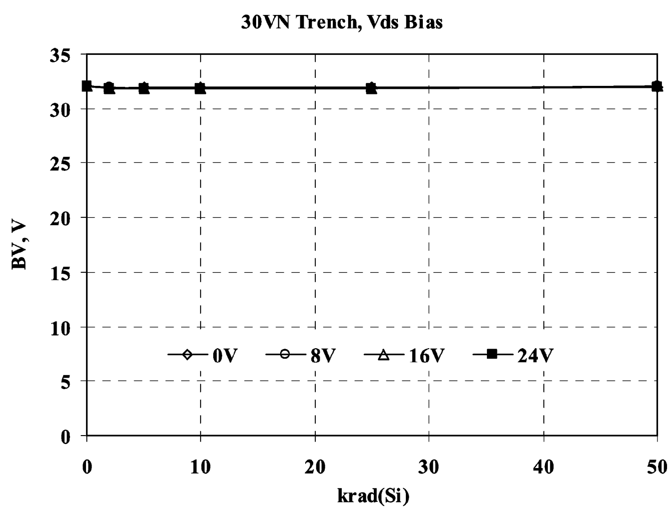

Fig. 12. The drain bias voltage under radiation showed little impact on breakdown voltage (BV) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

called reverse-biased gate leakage current—IgssR. Usually both IgssF and IgssR would be at only a few nano-ampers, but one can be significantly higher than the other due to different device design and test conditions.

Figs. 7 and 8 show how the in-situ gate and drain biases impact the forward-biased gate leakage current $(I_{\rm GSS}F)$. $I_{\rm GSS}F$ is observed to increase with either increasing values (negatively or positively) of in-situ gate bias. Negative in-situ gate biases induced larger increases in $I_{\rm GSS}F$. The in-situ drain bias is observed to have little impact upon $I_{\rm GSS}F$. The in-situ drain and gate biases show minimal impact upon the reverse-biased gate leakage current $(I_{\rm GSS}R)$. Those data are not shown here, since the observed changes are negligible.

C. Effects of In-Situ Gate and Drain Biases Upon Drain Leakage Current $(I_{\rm DSS})$

Fig. 9 shows the observed changes in $I_{\rm DSS}$ under different in-situ drain biases. $I_{\rm DSS}$ is measured with drain voltage at 80% of the device's rated drain breakdown voltage $({\rm BV}_{\rm DSS})$ and the gate grounded. Clearly, in-situ drain biases only caused minor increases in $I_{\rm DSS}$ with increasing dose. Fig. 10 shows the observed changes in $I_{\rm DSS}$ under different in-situ gate biases. The Idss data at ${\rm Vgs} = 8$ and 16 V for 50 krad(Si) have reached the compliance limit on the tester (1×10^7 nA).

D. Effects of In-Situ Gate and Drain Biases Upon Drain Breakdown Voltage $(BV_{\rm DSS})$

Figs. 11 and 12 show the impacts of in-situ gate and drain biases under radiation on ${\rm BV}_{\rm DSS}$, which is measured at a drain current of 1 mA and the gate grounded. Again, in-situ drain biases during irradiation show negligible changes in ${\rm BV}_{\rm DSS}$; whereas, in-situ gate biases have a significant impact upon ${\rm BV}_{\rm DSS}$. There are two observed responses: (1) at zero or negative in-situ gate biases, ${\rm BV}_{\rm DSS}$ increases with increasing levels of radiation dose and (2) at positive in-situ gate biases, ${\rm BV}_{\rm DSS}$ decreases with increasing levels of radiation dose. This second effect is observed because under positive in-situ biases, $V_{\rm TH}$ decreased significantly and in some cases shifted below zero volts. As $V_{\rm TH}$ shifts (towards zero or even negative), the

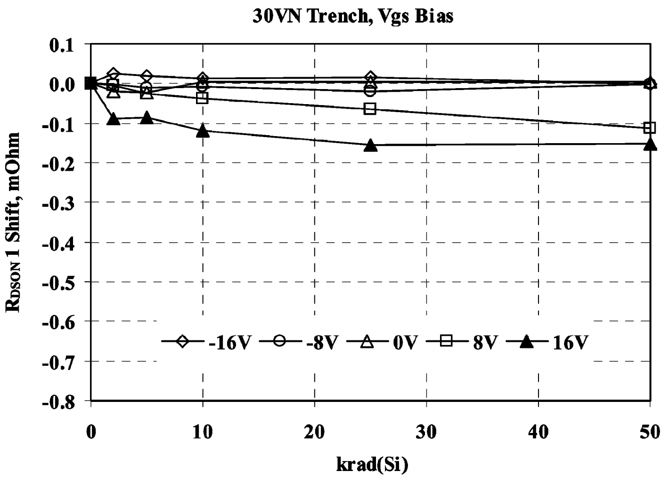

Fig. 13. The impact of gate bias voltage under radiation on Rdson1 (tested at $\mathrm{Vgs} = 10\ V$) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

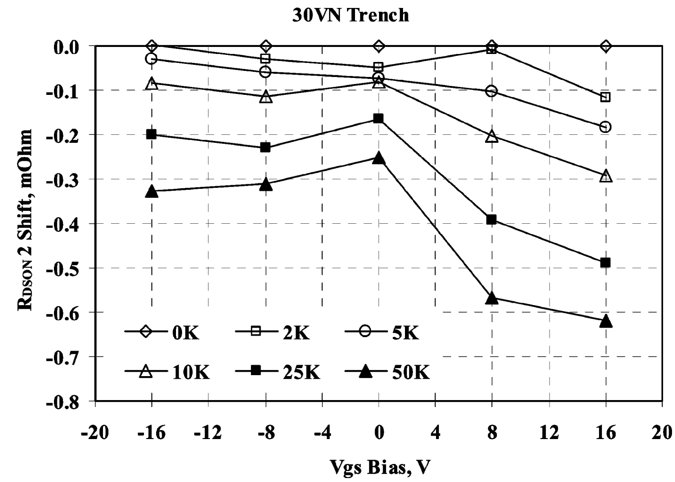

Fig. 15. The $R_{\mathrm{dson}}2$ shifts at different gate bias voltages under irradiation follow the threshold voltage shifts as shown in Fig. 6.

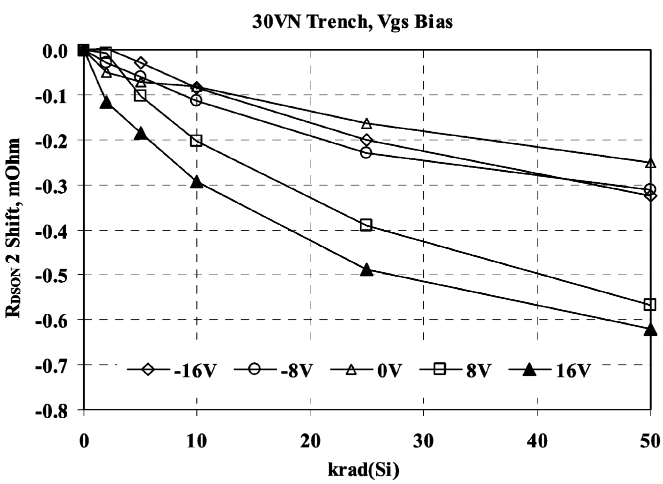

Fig. 14. The impact of gate bias voltage under radiation on $R_{\mathrm{dson}}2$ (tested at $\mathrm{Vgs} = 4.5\ V$) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

normally-off channel between the source and drain begins to conduct current (begins to form a conducting channel), which causes the observation of a decrease in $\mathrm{BV_{DSS}}$. If the V_{TH} shift actually forms a channel, the device is no longer capable of blocking a voltage. To alleviate this false measurement of breakdown voltage, one could reverse bias the gate, maintaining an off-state condition, and measure the breakdown voltage. An observation has been made that for the 30 V N channel trench power MOSFET, the breakdown voltage decreases with increasing negative gate bias even with no radiation [7]. It has been verified with both bench test and device simulation and it is due to trench device structure.

E. Effects of In-Situ Gate and Drain Biases Upon On-state Resistance (R_{DSON})

Figs. 13 and 14 show the impact of in-situ gate biases during irradiation upon the on-state resistances, $R_{\mathrm{DSON}}1$ and $R_{\mathrm{DSON}}2$. $R_{\mathrm{DSON}}1$ and $R_{\mathrm{DSON}}2$ are measured at a gate bias of 10 V and 4.5 V, respectively. After irradiation, $R_{\mathrm{DSON}}2$ decreases substantially when compared to $R_{\mathrm{DSON}}1$ and to its initial pre-rad value (with $\mathrm{DirectFET^{TM}}$ packages, the 30 VN trench device is nominally rated at 1.3 m Ω at 10 V and at 1.9 m Ω at 4.5 V). To understand the decrease observed in $R_{\mathrm{DSON}}2$ with increasing dose level, the data from Fig. 14 is re-plotted as shown in Fig. 15. Changes in $R_{\mathrm{DSON}}2$ follow the observed changes in V_{TH} as shown in Fig. 6. When V_{TH} becomes lower or negative due to trapped charge in the gate oxide, the channel becomes more inverted, which, in turn, lowers the channel resistance, making the measured R_{DSON} lower. Changes in $R_{\mathrm{DSON}}2$ under different in-situ drain biases also follow the observed changes in V_{TH} under those same in-situ bias conditions. In other words, when V_{TH} decreases, R_{DSON} at a given V_{GS} also decreases because the channel inversion voltage $(V_{\mathrm{GS}} - V_{\mathrm{TH}})$ increases as V_{TH} decreases.

F. Effects of In-Situ Gate and Drain Bias Upon Diode Forward Voltage (V_{SD})

Figs. 16 and 17 show the impact of in-situ drain and gate biases during irradiation upon the diode forward voltage (V_{SD}). In-situ drain biases had only a minor impact on V_{SD} (V_{SD} is observed to decrease approximately 50 mV after 50 krad(Si)), while in-situ gate biases (positive gate biases had the largest impact) caused a significant decrease in V_{SD} (V_{SD} is observed to decrease by approximately 450 mV after 50 krad(Si)).

To aid the understanding of why V_{SD} decreases with increasing dose level, the data used in Fig. 17 is re-plotted in Fig. 18. V_{SD} changes are minimal at low dose levels (i.e., 2, 5 and 10 krad(Si)) at all in-situ gate biases; V_{SD} changes increased significantly at higher dose levels (i.e., 25 and 50 krad(Si)). The pattern of V_{SD} changes also followed the changes in V_{TH} as shown in Fig. 6. V_{SD} is measured with gate-to-source shorted, and with a continuous source current I_{S}. Under normal device operation, I_{S} would normally flow from the source, through the body P-N junction, then to the drain. However, if the device is irradiated to higher dose levels with the gate positively biased, the threshold voltage of the device would reduce significantly or even become negative, channel resistance would become very small, then I_{S} would follow a path through this conductive channel to the drain, yielding a significantly lower V_{SD}

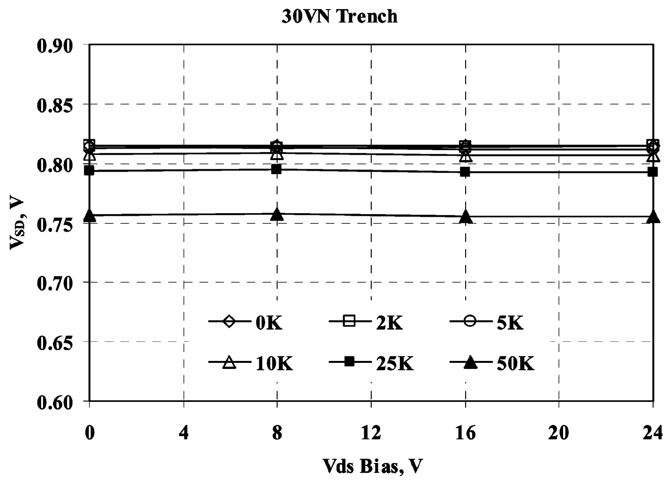

Fig. 16. The impact of drain bias voltage under radiation on forward body voltage drop ($V_{\rm SD}$) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

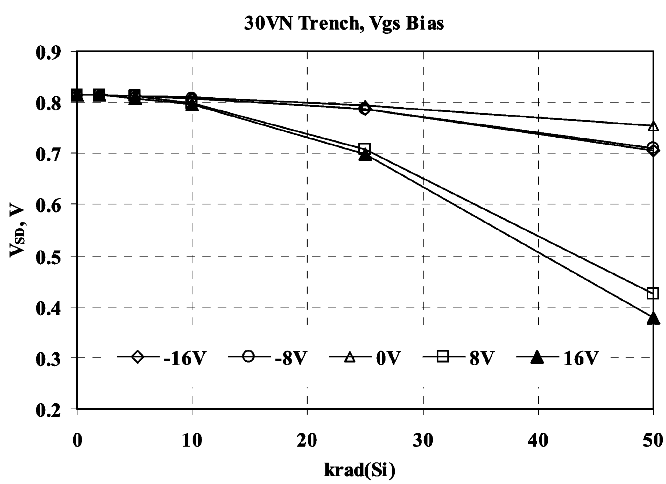

Fig. 17. The impact of gate bias voltage under radiation on forward body voltage drop ($V_{\rm SD}$) relative to total ionizing dose levels for a 30 V N channel trench power MOSFET.

measurement. At this point, the $V_{\rm SD}$ measurement no longer represents the body diode forward voltage, but the voltage drop when $I_{\rm S}$ flows from source, through the channel to the drain (voltage drop across the source, channel, and drain). To achieve the true $V_{\rm SD}$ value in such a situation, a negative voltage will be needed on the gate to keep channel off.

IV. Summary

Detailed analysis of total ionizing dose test results on commercial trench power MOSFETs show trench power MOSFETs have similar responses as radiation hardened power MOSFETs but with much larger shifts on most parameters. The electrical characteristics of commercial trench MOSFETs such as $V_{\rm TH}$,

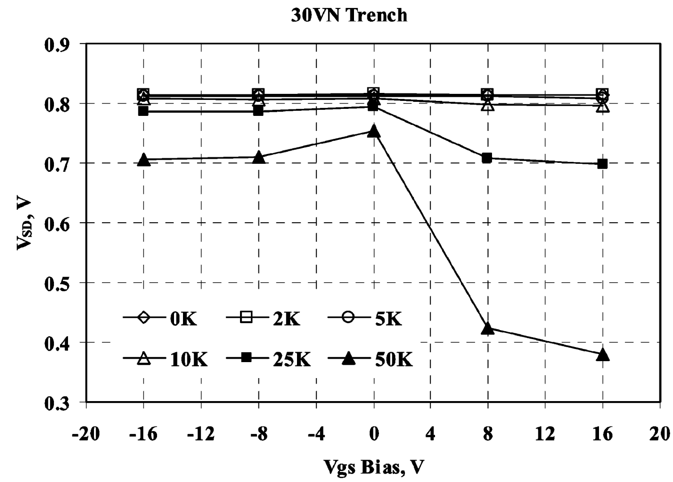

Fig. 18. The trend of $V_{\rm SD}$ shifts at higher total dose level follow the threshold voltage shifts as shown in Fig. 6.

$I_{\rm GSS}$, $I_{\rm DSS}$, $\rm BV_{DSS}$, $R_{\rm DSON}$, and $V_{\rm SD}$ will change significantly under certain bias conditions after approximately 25 krad(Si), but those changes have been attributed to trapped charge in the gate oxide allowing the normally-off channel to form a conductive path. Commercial power MOSFETs are not recommended to be used in space applications where significant radiations are to be encountered. The incorporation of a hardened gate oxide would mitigate many of the observed total dose effects observed on these commercial trench power MOSFETs.

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